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IDENTIFICATION OF AGE-DEPENDENT EFFECTS OF FUNDUS IMAGES ON INTRAOCULAR PRESSURE FOR GLAUCOMA PATIENTS: A TENSOR REGRESSION APPROACH

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Intraocular pressure (IOP) is a critical biomarker for the progression of glaucoma, which is often associated with elevated fluid pressure inside the eye. Sustained high IOP can impair the optic nerve over time, making early detection and continuous monitoring of IOP essential for glaucoma diagnosis, treatment, and disease management. A fundus image is a photograph of the back of the eye that allows doctors to see and evaluate the optic nerve head, retina, and blood vessels. Using such images to predict the IOP is a very relevant medical research topic in recent years. However, existing methods cannot capture the complicated age-dependent effects of image predictors. We thus propose a new varying coefficient model for tensor image data in this paper. We first employ a tensor partitioning strategy to reduce dimensionality, followed by a tensor decomposition technique for the high-dimensional tensor covariates. After extracting key features from the tensor covariates, we feed these low-dimensional representations into a varying coefficient model, alongside other one-way covariates. Additionally, we apply a non-concave penalty estimation to simultaneously identify the model structure and select significant predictors. A subsequent smoothing step enhances the model's accuracy. Extensive simulations are conducted to evaluate the performance of our approach. A real fundus image dataset is analyzed using our model to improve glaucoma management.

介绍青光眼的流行病学背景与预测的重要性，并强调基于眼部图像进行预测具有重要公共卫生意义。

1. Introduction. Glaucoma is a chronic eye condition that can lead to optic nerve damage and vision loss and a degenerative optic neuropathy characterized by the loss of retinal ganglion cells, often leading to irreversible blindness (Jonas et al. (2017)). It affects over 64.3 million individuals worldwide, with projections suggesting an increase to 111.8 million by 2040 (Tham et al. (2014)). This heterogeneous group of disorders is the second leading cause of blindness globally, potentially impacting up to 91 million people. Glaucoma presents a wide range of risk factors, including optic nerve head (ONH) damage, visual field (VF) loss, and elevated intraocular pressure (IOP). Since glaucoma can be asymptomatic in its early and intermediate stages, diagnosis often occurs only after irreversible optic nerve damage and vision loss have already happened. Typically, the progression of glaucoma is examined by experienced ophthalmologists through retinal assessments, a process that is both tedious and time-consuming. Therefore, predicting glaucoma accurately is a critical objective for ophthalmology researchers. Recent efforts have focused on leveraging eye images to enhance predictive accuracy, which could have a significant public health impact.

说明散瞳眼底摄影数据来源与研究数据集情况，并指出本文目标是基于眼底图像准确预测眼压。

Dilated fundus photography offers a convenient and inexpensive method for recording ONH structure, with glaucomatous optic neuropathy assessment remaining a gold standard

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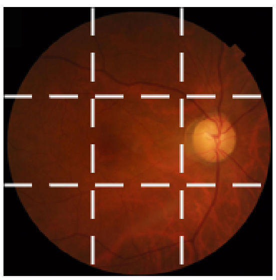
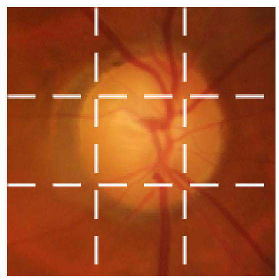
for glaucoma indication. See Figure 1 for examples of such images from the glaucoma real-world appraisal progression ensemble (GRAPE) dataset (Huang et al. (2023)) which we will analyze in this paper. The longitudinal image dataset was collected at the Eye Center of the Second Affiliated Hospital of Zhejiang University. It comprises 1,115 fundus images from 263 eyes, along with associated clinical information. In particular, increased IOP is the most important biomarker for glaucoma and medical researchers are especially interested in an accurate prediction of the IOP outcome based on fundus images. Establishing such predictive models is essential for enabling timely medical intervention, which in turn can slow or prevent the progression of glaucoma in at-risk individuals.

人工诊断和深度学习方法的局限性，并引出本文提出的张量建模方法作为替代方案。

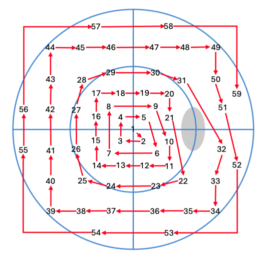
In practice, manual assessment of the optic disc in fundus photographs requires extensive clinical training, is highly subjective with limited specialist agreement, and is time-consuming. Recent advances in AI and deep learning, along with more available data, offer promise for objective glaucoma diagnosis. However, these methods typically need large, clinically annotated datasets and are often difficult to interpret. In this work, we propose a tensor modeling technique to address these challenges.



(a) CFP OD image with a tensor partition strategy



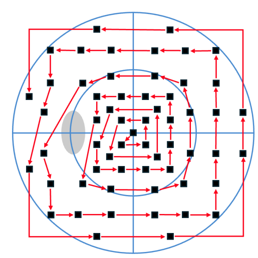
(b) ROI OD image with a tensor partition strategy



(c) VF OD schematic diagram and location IDs

(d) CFP OS image

(e) ROI OS image



(f) VF OS schematic diagram

Fig 1: Fundus image analysis: illustrations for different images in the GRAPE data set. (a) and (d): original color fundus photograph (CFP). (b) and (e): processed region of interest (ROI) image. (c) and (f): view field (VF) schematic diagram. Dashed lines in (a) and (b) show the proposed tensor partition on CFP and ROI images. The red arrowed curves in (c) and (f) show the extraction order of VF values of oculus dexter (OD) and oculus sinister(OS) respectively, and detailed location IDs reported in (c).

本段介绍了张量在高维多模态数据分析中的重要性及其在统计领域的主要研究方向。

Tensors, as multidimensional generalizations of matrices, have become increasingly prominent due to their ability to represent and analyze complex, high-dimensional data structures. Extending from low-dimensional vectors and matrices, tensor objects can encapsulate data across multiple dimensions, offering a more comprehensive representation of multiway data. This flexibility makes tensors particularly valuable in many fields, such as neuroimaging (Zhou, Li and Zhu (2013); Li et al. (2018)), genomics (Hore et al. (2016)), social network analysis (Dunlavy, Kolda and Acar (2011)), hyperspectral image (Liu et al. (2019)), and recommendation system (Zhang et al. (2021)). The inherent high-dimension in tensors (Hillar and Lim (2013)) pose difficulties and challenges in real data analysis. Recent studies in statistical tensors mainly focused on tensor decomposition (Kolda and Bader (2009); Zhang and Xia (2018); Zhang and Han (2019)), tensor completion (Gandy, Recht and Yamada (2011); Yuan and Zhang (2016); Zhang (2019)), tensor regression (Zhou, Li and Zhu (2013); Zhang and Li (2017); Li et al. (2018); Zhang et al. (2020)) and tensor clustering (Han et al. (2022); Luo and Zhang (2022)).

本段回顾了张量回归的主要模型类型及其通过低秩分解和稀疏正则化进行降维估计的主流方法。

We will consider eye images as input covariates to predict IOP using tensor regression methods. Previous authors extend traditional linear regression to tensor data, including scalar-on-tensor (Zhou, Li and Zhu (2013)), tensor-on-tensor (Lock (2018); Liu, Liu and Zhu (2020); Luo and Zhang (2022)), and tensor-on-vector (Li and Zhang (2017); Sun and Li (2017)), among others. In these problems, the number of unknown parameters can be enormous, often far exceeding the sample size, thus imposing a significant computational burden on the estimation. To address this challenge, dimension reduction tools are necessary for an efficient estimation of tensor coefficients, including the CP decomposition-based regression (Zhou, Li and Zhu (2013); Sun and Li (2017); Bi, Qu and Shen (2018); Hao, Zhang and Cheng (2020)), the Tucker decomposition-based regression (Li et al. (2018); Guhaniyogi, Qamar and Dunson (2017); Zhang et al. (2020); Han, Willett and Zhang (2022); Luo and Zhang (2023)), and the tensor train (TT) decomposition-based regression (Liu, Liu and Zhu (2020); Liu, Lee and Zhang (2024)), among others. In addition to aforementioned low rank representation methods, one may further impose a sparsity structure and use regularization in the estimation (Zhou, Li and Zhu (2013); Zhang et al. (2020); Liu, Liu and Zhu (2020)).

回顾了变系数模型的发展及其在处理非线性、动态与时空数据中的应用。

Most earlier works only examined a parametric linear relationship between tensors and other variables. To capture non-linear association, varying coefficient model (VCM) offers a more flexible framework (Hastie and Tibshirani (1993); Cai, Fan and Li (2000); Fan and Zhang (1999)). There exist a large body of literature for the development of estimation methods for VCM, such as the local linear method (Zhang, Lee and Song (2002); Xia, Zhang and Tong (2004); Fan (2018); Lu (2008)), the profile least squares method (Fan and Huang (2005)), the average derivatives method (Newey and Stoker (1993)), and the smoothing spline method (Ahmad, Leelahanon and Li (2005)). VCM is especially useful for addressing dynamic and complex data problems. Cheng et al. (2014) and Cheng, Honda and Li (2016) developed a comprehensive framework of VCM in longitudinal/cluster data. Mu, Wang and Wang (2018) applied a bivariate spline on triangulation to estimate the functional coefficient of space domain data. Yu et al. (2022) used tensor-product splines to evaluate the spatio-temporal non-stationarity. Zhu, Cai and Ma (2022) studied a kernel based varying coefficient network model.

本段指出现有研究尚未在张量回归中引入变系数结构，并提出本文的核心创新：张量回归的变系数模型及其在年龄图像交互预测中的应用。

From our review, very few papers introduced the nonparametric functional effect into tensor data analysis. Chen et al. (2024) proposed a semi-parametric tensor decomposition by assuming a constant core tensor and functional factor matrices generated from tucker decomposition. Han, Shi and Zhang (2023) developed an iterative algorithm to estimate functional factor matrix along one functional axis. Zhou, Wong and He (2024) investigated a nonparametric tensor regression model and used the spline approximation by the scale-adjusted block relaxation algorithm under CP decomposition. None of these works considered varying coefficients for tensor inputs. We propose a new VCM for tensor regression and study its estimation methods in this work. In particular, we may examine the complicated age-image interactions in the glaucoma management study and demonstrate the superior prediction performance of the proposed VCM.

概述模型的整体框架：

通过张量分块与 CP 分解降维，将问题转化为可用标准 VCM 方法估计的形式，并通过模拟与真实数据验证模型优势。

The varying-coefficient tensor regression model formulated in this paper can be viewed as a combination of tensor partition, tensor decomposition and varying coefficient partial linear model. Under our framework, we initially partition the whole tensor data into distinct, non-overlapping parts based on prior knowledge or distribution principles. The CP decomposition technique is applied to each partitioned tensor to extract low-rank factor representations. We only need to focus on the factor matrix along the subject dimension and with some linear algebra operation, we convert the high-dimensional functional coefficient tensor to a parsimonious functional coefficient matrix. Finally, standard nonparametric estimation methods of VCM can be implemented by vectorizing the transformed matrix covariate. Through extensive simulation studies we demonstrate the advantage of our proposed model. We finally apply this new model to examine the GRAPE data set and make some new findings for ophthalmology research.

The rest of this article is organized as follows. In Section 2, we introduce the methodology of our varying coefficient tensor regression model. In Section 3, we consider variable selection and model identification using a penalization estimation. In Section 4, simulation studies are conducted to examine the performance of the proposed model. In Section 5, we analyze the GRAPE data using the proposed model and report our findings. Theorem, technical proofs and additional experiment results are given in the Supplementary Material.

2. Methodology.

2.1. *Notations and Preliminaries.* We first review some basic mathematical facts on tensor (Kolda and Bader (2009)). In this paper, we use lowercase letters (e.g. x), lowercase boldface letters (e.g. $\mathbf{x}$), uppercase boldface letters (e.g. $\mathbf{X}$), and uppercase calligraphic letters (e.g. $\mathcal{X}$) to denote scalars, vectors, matrices, and order-3-or-higher tensors respectively. An order-D **tensor** $\mathcal{X} = (x_{j_1 j_2 \cdots j_D}) \in \mathbb{R}^{p_1 \times p_2 \times \cdots \times p_D}$, $j_d = 1, \cdots, p_d$, $d = 1, \cdots, D$, is a D-dimensional array with $\Pi_{d=1}^{D} p_d$ elements. A vector or a matrix may be regarded as order-1 or order-2 tensor respectively. We denote the **inner product** of two tensors $\mathcal{X}, \mathcal{Y} \in \mathbb{R}^{p_1 \times p_2 \times \cdots \times p_D}$ as

“同一个 index 位置”的元素相乘，再全部相加

$$(1) \qquad \langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{j_1=1}^{p_1} \sum_{j_2=1}^{p_2} \cdots \sum_{j_D=1}^{p_D} x_{j_1, j_2, \cdots, j_D} y_{j_1, j_2, \cdots, j_D}$$

An **outer product** for D vectors $\mathbf{x}^{(d)} \in \mathbb{R}^{p_d}$, $d = 1, 2, \cdots, D$, is an order-D tensor denoted as $\mathcal{X} = \mathbf{x}^{(1)} \circ \mathbf{x}^{(2)} \circ \cdots \circ \mathbf{x}^{(D)}$, and such $\mathcal{X}$ is called rank-1 tensor.

一个D-dim矩阵，[p1,...,pd]坐标元素，就是对应向量里的元素全部乘起来。

The parallel factor analysis, also known as **CP decomposition**, factorizes a tensor into a linear combination of rank-1 tensors. In particular, a tensor satisfying a rank R CP decomposition has the following structure

不是对角化，而是用R个rank-1 tensor去表达原张量

$$(2) \qquad \mathcal{X} = \sum_{r=1}^{R} \lambda_r \mathbf{x}_r^{(1)} \circ \mathbf{x}_r^{(2)} \circ \cdots \circ \mathbf{x}_r^{(D)}$$

where $\mathbf{x}_r^{(d)} = (x_{j_d r}^{(d)}) \in \mathbb{R}^{p_d}$, $j_d = 1, \cdots, p_d$, $d = 1, \cdots, D$, $r = 1, \cdots, R$, are normalized rank-1 tensor, and $\lambda_1 \geq \cdots \geq \lambda_R \geq 0$ are the weights.

In this paper, suppose we observe data $\{(y_{ij}, \mathcal{X}_{ij}, \mathbf{z}_i, t_i) : i = 1, \cdots, n, \ j = 1, 2\}$ from a sample of n units, where for the i-th unit, $y_{ij} \in \mathbb{R}$ is the response for the jth eye, $\mathcal{X}_{ij} = (x_{ij, j_1 \cdots j_D}) \in \mathbb{R}^{p_1 \times \cdots \times p_D}$ is an order-D tensor covariate, $\mathbf{z}_i = (z_{i,1}, \cdots, z_{i,p_0})^\top \in \mathbb{R}^{p_0}$ is a p_0-vector of scalar covariates, and $t_i \in [0, 1]$ is an index variable.

图像 X[i,j] 有一组回归系数 A[ti]， 其中 A
会随着患者年龄改变，所以有 [ti]

2.2. *Tensor partition varying-coefficient model.* Our interest is to develop a regression model for the association between the scalar response y, and its corresponding tensor covariate $\mathcal{X}$ and one-way covariates $\mathbf{z}$. We shall assume that y, $\mathcal{X}$ and $\mathbf{z}$ are centered. The regression coefficients for the tensor covariate are allowed to vary with an index variable $t \in [0,1]$. Specifically, for the i-th unit, we assume

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}. \tag{3}$$

In model (3), the effect of tensor covariate on response is treated as varying tensor function $\mathcal{A}(t_i) = (\alpha_{j_1,\cdots,j_D}(t_i))$, where each $\alpha_{j_1,\cdots,j_D}(t_i) : [0,1] \to \mathbb{R}$ is an unknown mapping. The coefficient for one-way covariate $\mathbf{z}_i$ is a constant vector $\boldsymbol{\beta} \in \mathbb{R}^{p_0}$. In addition, we assume $(\epsilon_{i1}, \epsilon_{i2})^T$ are i.i.d. from a bivariate normal distribution with mean zero and covariance $\Sigma_0 = \begin{pmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{pmatrix}$. In this model, the index variable t_i can be any continuous covariate. For example, in the glaucoma study, investigators are interested in examining how the effects of fundus images on the intraocular pressure change with age and we thus choose t to be the patient's age in this study.

A commonly observed image tensor $\mathcal{X}$ would require the estimation of enormous unknown varying-coefficients. It is computationally prohibitive to directly estimate $\mathcal{A}(t)$ due to the high dimensionality. However, not all elements of $\mathcal{X}$ contribute to the response of interest. Often, regions of interest are confined to a small portion of the entire tensor. Therefore, it is more sensible to focus on sub-tensors that capture those local features. To this end, we consider a partition model that divides the concatenated full-sample tensor $\tilde{\mathcal{X}}$ into S disjoint sub-tensor covariates $\tilde{\mathcal{X}}^{(s)}$, that is

$$\tilde{\mathcal{X}} = \cup_{s=1}^S \tilde{\mathcal{X}}^{(s)} \quad \text{and} \quad \tilde{\mathcal{X}}^{(s)} \cup \tilde{\mathcal{X}}^{(s')} = \emptyset \quad \forall s \neq s'. \tag{4}$$

Without loss of generality, we assume that the size of each sub-tensor $\tilde{\mathcal{X}}^{(s)} \in \mathbb{R}^{n \times p_1' \times \cdots \times p_D'}$ is fixed for any s such that $S = \Pi_{d=1}^D (p_d/p_d') \equiv \Pi_{d=1}^D S^{(d)}$, where $S^{(d)}$ is the number of partitions along the d-th order. We denote $\mathcal{X}_i^{(s)} \in \mathbb{R}^{p_1' \times \cdots \times p_D'}$ as partition s of the i-th tensor 第[i]个人的块s
covariate accordingly, and $\tilde{\mathcal{X}}^{(s)} = \{\mathcal{X}_i^{(s)} : i = 1, \cdots, n\}$. 所有人的块s

Corresponding to the partition pattern of $\tilde{\mathcal{X}}$, we can also divide $\mathcal{A}(t)$ into S disjoint sub-tensor functions $\mathcal{A}^{(s)}(t)$, that is

$$\mathcal{A}(t) = \cup_{s=1}^S \mathcal{A}^{(s)}(t) \quad \text{and} \quad \mathcal{A}^{(s)}(t) \cup \mathcal{A}^{(s')}(t) = \emptyset \quad \forall s \neq s'. \tag{5}$$

With such tensor partition technique, we can rewrite the tensor effect in (3) as $\langle \mathcal{X}_i, \mathcal{A}(t_i) \rangle = \sum_{s=1}^S \langle \mathcal{X}_i^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle$.

Next we adopt a rank-R CP decomposition model on each partition to further reduce the sub-tensor $\tilde{\mathcal{X}}^{(s)}$ to lower dimensional feature matrices. Specifically, we have

因为n个样本拼在一起了，
所以多了一维，长度为n

$$\tilde{\mathcal{X}}^{(s)} \approx \sum_{r=1}^R \lambda_r^{(s)} \mathbf{x}_{rs}^* \circ \mathbf{x}_r^{(s1)} \circ \cdots \circ \mathbf{x}_r^{(sD)}, \tag{6}$$

where $\lambda_r^{(s)}$ is the weight of the r-th rank of the decomposition, $\mathbf{x}_r^{(sd)} = (x_{1r}^{(sd)}, \cdots, x_{p_d' r}^{(sd)})^\top \in \mathbb{R}^{p_d'}$ is the common factor along the d-th order of the r-th rank, and $\mathbf{x}_{rs}^* = (x_{1rs}^*, \cdots, x_{nrs}^*)^\top \in \mathbb{R}^n$ is the factor along the subject dimension. The elements of $\mathbf{x}_{rs}^*$ contain the primary variation in the tensor variables due to subject differences, while the common structure among the subjects is absorbed into factors $\mathbf{x}_r^{(sd)}$, $r = 1, \cdots, R$, $d = 1, \cdots, D$ (Miranda et al. (2018)). If we focus on the i-th individual tensor covariate $\mathcal{X}_i$, following (6), we have

每一个人可以单拎出来

$$\mathcal{X}_i^{(s)} \approx \sum_{r=1}^R \lambda_r^{(s)} x_{irs}^* \mathbf{x}_r^{(s1)} \circ \cdots \circ \mathbf{x}_r^{(sD)} \equiv \sum_{r=1}^R x_{irs}^* \mathcal{U}_r^{(s)}, \tag{7}$$

where $\mathcal{U}_r^{(s)} = \lambda_r^{(s)} \mathbf{x}_r^{(s1)} \circ \cdots \circ \mathbf{x}_r^{(sD)} \in \mathbb{R}^{p_1' \times \cdots \times p_D'}$. We note that $\mathcal{U}_r^{(s)}$ contains only the shared information among subjects and thus can be dropped from the regression analysis.

After the above partitioning and decomposition, essentially only x_{irs}^* may contribute to the response variation and thus serve as an appropriate covariate. Consequently, the total number of measurements for an individual tensor is reduced from $p_1 \times \cdots \times p_D$ to $R \times S$. A similar method is adopted in Tang, Bi and Qu (2020) to learn individualized tensor construction. If we denote $a_{rs}(t) \equiv \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \rangle$ as the corresponding coefficient function of x_{irs}^*, $i = 1, \cdots, n$, and $\mathbf{A}(t) = (a_{rs}(t))$, model (3) now becomes

$$
\begin{aligned}
y_{ij} =& \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\
=& \sum_{s=1}^{S} \langle \mathcal{X}_{ij}^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\
\approx& \sum_{s=1}^{S} \left\langle \sum_{r=1}^{R} x_{ijrs}^* \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\
=& \sum_{s=1}^{S} \sum_{r=1}^{R} x_{ijrs}^* \underline{\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle} + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\
=& \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij},
\end{aligned}
\tag{8}
$$

where $\mathbf{X}_{ij}^* = (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$. Throughout this paper, we shall assume that R and S are constants. Instead of applying the decomposition on coefficient tensor $\mathcal{A}$ (Zhou, Li and Zhu (2013); Zhang and Li (2017); Li et al. (2018)), we place the low-rank assumption on the tensor covariate $\mathcal{X}$ and make no specific assumptions regarding the structure of the tensor function $\mathcal{A}(t)$. Furthermore, we do not need more assumptions on tensor covariate $\mathcal{X}$, such as Gaussian ensemble design (Zhang et al. (2020); Han, Willett and Zhang (2022)) or tensor restricted isometry property (Luo and Zhang (2023); Tong et al. (2022)). The model considered in this paper thus presents a novel introduction to the literature.

2.3. *Estimating functions and parameters.* For the unknown functional coefficients, we only require the usual smoothness condition which enables the local linear approximation by a first order Taylor expansion. Specifically, for t_i is in a small neighborhood of t, we have

$$
\mathbf{A}(t_i) \approx \mathbf{A}(t) + \dot{\mathbf{A}}(t)(t_i - t),
\tag{9}
$$

where $\dot{\mathbf{A}}(t)$ is the first derivative of $\mathbf{A}(t)$. We may estimate the unknown functions $\mathbf{A}(t)$ and parameters $\boldsymbol{\beta}$ by minimizing the local least squares

$$
\begin{aligned}
&\sum_{i=1}^{n} \sum_{j=1}^{2} \left[y_{ij} - \mathbf{v}_{ij}^\top (\mathsf{a}_t + \mathsf{b}_t (t_i - t)) - \mathbf{z}_i^\top \boldsymbol{\beta} \right]^2 K_h(t_i - t) \\
=& \sum_{i=1}^{n} \sum_{j=1}^{2} \left[y_{ij} - \tilde{\mathbf{v}}_{ij}^\top(t) \boldsymbol{\theta}_t \right]^2 K_h(t_i - t),
\end{aligned}
\tag{10}
$$

β 在那里只是“顺带估计”的 nuisance parameter。

纯数据矩阵！与未知参数无关，

where $\mathbf{v}_{ij} = \mathrm{vec}(\mathbf{X}_{ij}^*) \in \mathbb{R}^{RS}$, $\mathsf{a}_t = \mathrm{vec}(\mathbf{A}(t))$, $\mathsf{b}_t = \mathrm{vec}(\dot{\mathbf{A}}(t))$, $\tilde{\mathbf{v}}_{ij}^\top(t) = (\mathbf{z}_i^\top, \mathbf{v}_{ij}^\top, (t_i - t)\mathbf{v}_{ij}^\top)^\top \in \mathbb{R}^{2RS+p_0}$, $\boldsymbol{\theta}_t = (\boldsymbol{\beta}^\top, \mathsf{a}_t^\top, \mathsf{b}_t^\top)^\top$ and $K_h(t) = K(t/h)/h$ is a scaled kernel with bandwidth h. It is straightforward to derive the closed-form solution of (10) to be

$$
\hat{\boldsymbol{\theta}}_t^\dagger = \left[\tilde{\mathbf{V}}^\top(t) \mathbf{W}(t) \tilde{\mathbf{V}}(t) \right]^{-1} \tilde{\mathbf{V}}^\top(t) \mathbf{W}(t) \mathbf{y},
\tag{11}
$$

所有待估参数

where $\mathbf{y} = (y_{11}, y_{12} \cdots, y_{n1}, y_{n2})^\top \in \mathbb{R}^{2n}$, $\tilde{\mathbf{V}}(t) = (\tilde{\mathbf{v}}_{11}(t), \cdots, \tilde{\mathbf{v}}_{n2}(t))^\top \in \mathbb{R}^{2n\times(2RS+p_0)}$, and $\mathbf{W}(t) = \mathrm{diag}(K_h(t_1 - t), K_h(t_1 - t), \cdots, K_h(t_n - t)) \in \mathbb{R}^{2n\times 2n}$. We thus have vectorized estimate $\mathrm{vec}(\hat{\mathbf{A}}^\dagger(t)) = \hat{\mathrm{a}}_t^\dagger = (\mathbf{0}_{RS\times p_0}, \mathbf{I}_{RS}, \mathbf{0}_{RS\times RS})\hat{\boldsymbol{\theta}}_t^\dagger$. In practice, the bandwidth h is chosen by the leave-one-out cross validation.

After the function estimate $\hat{\mathbf{A}}^\dagger(t)$ is attained, we may evaluate the global estimator of $\boldsymbol{\beta}$ by minimizing

$$\sum_{i=1}^{n}\sum_{j=1}^{2}\left[y_{ij} - \langle \mathbf{X}_{ij}^*, \hat{\mathbf{A}}^\dagger(t_i)\rangle - \mathbf{z}_i^\top \boldsymbol{\beta}\right]^2. \tag{12}$$

全局重新估计β

The estimate $\hat{\boldsymbol{\beta}}^\dagger$ is given by

$$\hat{\boldsymbol{\beta}}^\dagger = \left[\mathbf{Z}^\top \mathbf{Z}\right]^{-1}\mathbf{Z}\mathbf{y}^\dagger, \tag{13}$$

where $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_1, \cdots, \mathbf{z}_n)^\top \in \mathbb{R}^{2n\times p_0}$, and $\mathbf{y}^\dagger = (y_{11} - \langle \mathbf{X}_{11}^*, \hat{\mathbf{A}}^\dagger(t_1)\rangle, \cdots, y_{n2} - \langle \mathbf{X}_{n2}^*, \hat{\mathbf{A}}^\dagger(t_n)\rangle)^\top \in \mathbb{R}^{2n}$.

两个眼睛之间的corr怎么估计

However, the above estimators ignore the correlation between observations from the two eyes. We next consider a weighted estimator to improve the efficiency. From classical longitudinal data analysis, we can attain refined estimators to be

$$\hat{\boldsymbol{\theta}}_t^* = \left[\tilde{\mathbf{V}}^\top(t)\Sigma^{-1/2}\mathbf{W}(t)\Sigma^{-1/2}\tilde{\mathbf{V}}(t)\right]^{-1}\tilde{\mathbf{V}}^\top(t)\Sigma^{-1/2}\mathbf{W}(t)\Sigma^{-1/2}\mathbf{y}, \tag{14}$$

and

$$\hat{\boldsymbol{\beta}}^* = \left[\mathbf{Z}^\top\Sigma^{-1}\mathbf{Z}\right]^{-1}\mathbf{Z}\Sigma^{-1}\mathbf{y}^\dagger, \tag{15}$$

单个人是Σ_0，而Σ指单位阵的1换成Σ_0

if the covariance $\Sigma = \mathbf{I}_n \otimes \Sigma_0$ is known. To this end, we may estimate the covariance parameters by

$$\hat{\sigma}^2 = (2n)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{2}\left[y_{ij} - \langle \mathbf{X}_{ij}^*, \hat{\mathbf{A}}^\dagger(t_i)\rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^\dagger\right]^2, \tag{16}$$

and

$$\hat{\rho} = n^{-1}\sum_{i=1}^{n}\left[y_{i1} - \langle \mathbf{X}_{i1}^*, \hat{\mathbf{A}}^\dagger(t_i)\rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^\dagger\right]\left[y_{i2} - \langle \mathbf{X}_{i2}^*, \hat{\mathbf{A}}^\dagger(t_i)\rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^\dagger\right]/\hat{\sigma}^2. \tag{17}$$

We thus replace σ^2 and ρ in (14) and (15) by (16) and (17) and obtain the final estimator $\hat{\boldsymbol{\theta}}_t$ and $\hat{\boldsymbol{\beta}}$.

Remark 1 Under conditions (C1)–(C4) in the Supplementary, we show in Section C.1 that the local linear estimators for the time-varying coefficient matrix $A(t)$ and its derivative are jointly asymptotically normal with standard kernel-smoothing bias and variance of order $(nh)^{-1}$; the optimal bandwidth for estimating $A(t)$ is $h \sim n^{-1/5}$ (c.f. Theorem 1 in Section C.1). To achieve $\sqrt{n}$-consistency and asymptotic normality for the parametric component β, undersmoothing is required (i.e., choosing h to be smaller than $n^{-1/5}$ so that $nh^4 \to 0$ and $nh^3/\log n \to \infty$, c.f. Theorem 2 in Section C.1).

3. Variable selection and structure identification.

3.1. *Group penalty procedure.* Even after tensor partition and decomposition, there are still RS functions to be estimated in $\mathbf{A}(t) = (a_{rs}(t))$, $r = 1, \cdots, R$, $s = 1, \cdots, S$, in addition to p_0 Euclidean parameters in $\boldsymbol{\beta}$. The dimension may still be quite large relative to the available sample size. In practice not all these components are effective to the response and a sparsity structure is usually plausible. Furthermore, not all functional coefficients are varying and need to be estimated nonparametrically. We further adopt a regularization procedure for variable selection and model structure identification.

To implement the shrinkage approach, we consider spline basis approximation method (De Boor (2001)) for functional estimation in this section. While the kernel method can usually provide a better local approximation to the function estimation, it is often less efficient suitable for the penalized computations. In contrast, the spline method is computationally more efficient for our purpose of regularization.

For a chosen set of B-splines basis functions $\{B_l(t), l = 1, \cdots, L\}$, we write $\mathbf{B}(t) = (B_1(t), \cdots, B_L(t))^T$ where each $B_l(t)$ is a B-spline function, $L = k_n + k_d + 1$ is the number of basis functions, k_n is the number of equispaced internal knots located in $[0, 1]$, and k_d is the degree of the polynomial spline. The spline approximation of the varying coefficient can be written in terms of B-splines as

a[rs]的待估就变成了gamma系数的待估

$$a_{rs}(t) \approx \sum_{l=1}^{L} \gamma_{lrs} B_l(t) = \mathbf{B}^\top(t) \boldsymbol{\gamma}_{rs}, \tag{18}$$

where $\boldsymbol{\gamma}_{rs} = (\gamma_{1rs}, \cdots, \gamma_{Lrs})^T$ are some suitable coefficients.

Here we consider B-spline basis for its desirable properties such as the numerical stability (De Boor (2001)) while other basis functions can be used for a similar purpose. We need to choose k_n to control the smoothness of nonparametric function $a_{rs}(t)$'s and require $k_n \to \infty$ as $n \to \infty$. Although it is possible to allow different number of basis for different functions $a_{rs}(t)$, we specify them all equal to L for the simplicity of notation.

After substituting (18) into (8), we obtain a slightly different tensor regression form:

所以我们的模型长这样：

只剩下完成γ β 的估计

$$y_{ij} \approx \sum_{l=1}^{L} \sum_{r=1}^{R} \sum_{s=1}^{S} x^*_{ijrs} B_l(t_i) \gamma_{lrs} + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \equiv \langle \mathcal{H}_{ij}(t_i), \mathcal{G} \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}, \tag{19}$$

where $\mathcal{H}_{ij}(t) = \mathbf{X}^*_{ij} \circ \mathbf{B}(t) = (x^*_{ijrs} B_l(t)) \equiv (h_{ijlrs}(t)) \in \mathbb{R}^{L \times R \times S}$, and $\mathcal{G} = (\gamma_{lrs}) \in \mathbb{R}^{L \times R \times S}$.

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Before we define the penalty term in a projected B-spline space for simultaneous variable selection and structure identification, we introduce an orthogonal decomposition of $a_{rs}(t)$ with respect to the L_2 norm by

$$a_{rs}(t) = (a_{rs})_c + (a_{rs})_v(t), \tag{20}$$

where $(a_{rs})_c = \int_0^1 a_{rs}(t)dt \approx \int_0^1 \mathbf{B}^\top(t)dt \boldsymbol{\gamma}_{rs} \in \mathbb{R}$ is the mean of $a_{rs}(t)$, and $(a_{rs})_v(t) = a_{rs}(t) - (a_{rs})_c \approx \big(\mathbf{B}^\top(t) - \int_0^1 \mathbf{B}^\top(t)dt\big)\boldsymbol{\gamma}_{rs}$ is the deviation from the mean. Then we have

$$\|a_{rs}\|_2^2 = |(a_{rs})_c|^2 + \|(a_{rs})_v\|_2^2, \tag{21}$$

where $|\cdot|$ denotes the absolute value, and $\|\cdot\|_2$ is the L_2 norm. The two terms on the right hand side of (21) can be easily evaluated by $|(a_{rs})_c|^2 \approx \boldsymbol{\gamma}_{rs}^\top \int_0^1 \mathbf{B}(t)dt \int_0^1 \mathbf{B}^\top(t)dt \boldsymbol{\gamma}_{rs}$, and $\|(a_{rs})_v\|_2^2 \approx \boldsymbol{\gamma}_{rs}^\top \big[\int_0^1 \big(\mathbf{B}(t) - \int_0^1 \mathbf{B}(t)dt\big)\big(\mathbf{B}^\top(t) - \int_0^1 \mathbf{B}^\top(t)dt\big)dt\big]\boldsymbol{\gamma}_{rs} = \boldsymbol{\gamma}_{rs}^\top\big(\int_0^1 \mathbf{B}(t)\mathbf{B}^\top(t)dt - \int_0^1 \mathbf{B}(t)dt \int_0^1 \mathbf{B}^\top(t)dt\big)\boldsymbol{\gamma}_{rs}$. It is clear that $\|(a_{rs})_v\|_2 \neq 0$ when a_{rs} is a varying function, $\|(a_{rs})_v\|_2 = 0$ when a_{rs} is a constant function, and furthermore, $|(a_{rs})_c| = \|(a_{rs})_v\|_2 = 0$ when a_{rs} is a constant zero function.

We then propose the following penalized loss function to simultaneously select important variables and identify the model structure with group regularized estimator $\hat{\mathcal{G}}^{\ddagger}$ and $\hat{\boldsymbol{\beta}}^{\ddagger}$:

与B没关系，只与gamma（待估参数）有关

$$\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{2}\left[y_{ij}-\langle\mathcal{H}_{ij}(t_i),\mathcal{G}\rangle-\mathbf{z}_i^{\top}\boldsymbol{\beta}\right]^2+n\sum_{k=1}^{p_0}P_{\omega_\beta}(|\beta_k|)$$
$$+n\sum_{r=1}^{R}\sum_{s=1}^{S}\left\{P_{\omega_\gamma^c}(|(a_{rs})_c|)+P_{\omega_\gamma^v}(\|(a_{rs})_v\|_2)\right\}, \tag{22}$$

where $P_\omega(\cdot)$'s are all SCAD penalty functions with a group tuning parameter ω. In general the SCAD function is defined through its first derivative as

$$\dot{P}_\omega(\theta)=\omega\Big[I\{|\theta|\le\omega\}+\frac{(a_0\omega-|\theta|)_+}{(a_0-1)\omega}I\{|\theta|>\omega\}\Big] \tag{23}$$

with $a_0>2$, where $I\{\cdot\}$ is an indicator function. In our numerical studies, we set $a_0=3.7$ as suggested by Fan and Li (2001).

To implement the above minimization, following Fan and Li (2001), we use an iterative local quadratic approximation algorithm to solve the penalized estimation. Using the Taylor expansion, given an estimator of θ in $(c-1)$-th iteration, $\theta^{(c-1)}$, we can approximate the regularization term $P_\omega(\theta)$ by

$$P_\omega(\theta)\approx P_\omega(\theta^{(c-1)})+\frac{1}{2}\frac{\dot{P}_\omega(\theta^{(c-1)})}{\theta^{(c-1)}}(\theta^2-(\theta^{(c-1)})^2). \tag{24}$$

The initial estimate $\mathcal{G}^{(0)}$ and $\boldsymbol{\beta}^{(0)}$ can be obtained from $\mathbf{A}^{\dagger}(t)$ and $\boldsymbol{\beta}^{\dagger}$ in (11) and (13), or the least square estimation of (22) with $\omega_\beta=\omega_\gamma^c=\omega_\gamma^v=0$.

如何迭代完成γ β 的估计?

1. 先算β

In the c-th iteration, we first update $\boldsymbol{\beta}^{(c)}$ by minimizing

$$\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{2}\left[y_{ij}-\langle\mathcal{H}_{ij}(t_i),\mathcal{G}^{(c-1)}\rangle-\mathbf{z}_i^{\top}\boldsymbol{\beta}\right]^2+n\sum_{k=1}^{p_0}P_{\omega_\beta}(|\beta_k|) \tag{25}$$

where $\mathcal{G}^{(c-1)}$ is the estimate of $\mathcal{G}$ in the $(c-1)$-th iteration. The approximate solution of (25) is

$$\boldsymbol{\beta}^{(c)}=\Big(\mathbf{Z}^{\top}\mathbf{Z}+n\boldsymbol{\Omega}_0(\boldsymbol{\beta}^{(c-1)},\omega_\beta)\Big)^{-1}\mathbf{Z}^{\top}\mathbf{y}_{\mathbf{Z}}^{(c-1)}, \tag{26}$$

where $\mathbf{y}_{\mathbf{Z}}^{(c-1)}=(y_{11}-\langle\mathcal{H}_{11}(t_1),\mathcal{G}^{(c-1)}\rangle,\cdots,y_{n2}-\langle\mathcal{H}_{n2}(t_n),\mathcal{G}^{(c-1)}\rangle)^{\top}\in\mathbb{R}^{2n}$, $\mathbf{Z}=(\mathbf{z}_1,\cdots,\mathbf{z}_n)^{\top}\in\mathbb{R}^{2n\times p_0}$, $\boldsymbol{\Omega}_0(\boldsymbol{\beta}^{(c-1)},\omega_\beta)=\mathrm{diag}\Big(\frac{\dot{P}_{\omega_\beta}(|\beta_1^{(c-1)}|)}{|\beta_1^{(c-1)}|},\cdots,\frac{\dot{P}_{\omega_\beta}(|\beta_{p_0}^{(c-1)}|)}{|\beta_{p_0}^{(c-1)}|}\Big)\in\mathbb{R}^{p_0\times p_0}$ is a diagonal matrix, and $\boldsymbol{\beta}^{(c-1)}=(\beta_1^{(c-1)},\cdots,\beta_{p_0}^{(c-1)})\in\mathbb{R}^{p_0}$ is the estimate of $\boldsymbol{\beta}$ in the $(c-1)$-th iteration.

如何迭代完成γ β 的估计?

2. 再算γ

Then, given the current estimate $\boldsymbol{\beta}^{(c)}$, we update the estimate $\mathcal{G}^{(c)}$ by minimizing

$$\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left[y_{ij}-\langle\mathcal{H}_{ij}(t_i),\mathcal{G}\rangle-\mathbf{z}_i^{\top}\boldsymbol{\beta}^{(c)}\right]^2+n\sum_{r=1}^{R}\sum_{s=1}^{S}P_{\omega_\gamma^c}(|(a_{rs})_c|)+n\sum_{r=1}^{R}\sum_{s=1}^{S}P_{\omega_\gamma^v}(\|(a_{rs})_v\|_2). \tag{27}$$

Similar to (26), $\mathcal{G}^{(c)}$ can be obtained through

$$\mathrm{vec}(\mathcal{G}^{(c)})=\Big(\mathbf{H}^{\top}\mathbf{H}+n\boldsymbol{\Omega}_1(\mathcal{G}^{(c-1)},\omega_\gamma^c)+n\boldsymbol{\Omega}_2(\mathcal{G}^{(c-1)},\omega_\gamma^v)\Big)^{-1}\mathbf{H}^{\top}\mathbf{y}_{\mathbf{H}}^{(c)}, \tag{28}$$

where $\mathbf{y}_{\mathbf{H}}^{(c)} = (y_{11} - \mathbf{z}_1^\top \boldsymbol{\beta}^{(c)}, \cdots, y_{n2} - \mathbf{z}_n^\top \boldsymbol{\beta}^{(c)})^\top \in \mathbb{R}^{2n}$, $\mathbf{H} = (\mathrm{vec}(\mathcal{H}_{11}(t_1), \cdots, \mathrm{vec}(\mathcal{H}_{n2}(t_n))^\top \in \mathbb{R}^{2n \times LRS}$, and $\mathrm{vec}(\mathcal{H}_{ij}(t_i)) = (h_{ijlrs}(t_i)) = (h_{ij111}(t_i), \cdots, h_{ijL11}(t_i), \cdots, h_{ijLR1}(t_i), \cdots, h_{ijLRS}(t_i))^\top \in \mathbb{R}^{LRS}$. As for two penalized terms, we have

$$\boldsymbol{\Omega}_1(\mathcal{G}^{(c-1)}, \omega_\gamma^c) = \mathrm{diag}\Big(\frac{\dot{P}_{\omega_\gamma^c}(|(a_{11}^{(c-1)})_c|)}{|(a_{11}^{(c-1)})_c|}\mathcal{B}_1, \cdots, \frac{\dot{P}_{\omega_\gamma^c}(|(a_{R1}^{(c-1)})_c|)}{|(a_{R1}^{(c-1)})_c|}\mathcal{B}_1, \cdots, \frac{\dot{P}_{\omega_\gamma^c}(|(a_{RS}^{(c-1)})_c|)}{|(a_{RS}^{(c-1)})_c|}\mathcal{B}_1\Big)$$

is an LRS-by-LRS block diagonal matrix, where $|(a_{rs}^{(c-1)})_c| = \sqrt{\boldsymbol{\gamma}_{rs}^{(c-1)\top}\mathcal{B}_1\boldsymbol{\gamma}_{rs}^{(c-1)}}$ with $\boldsymbol{\gamma}_{rs} = (\gamma_{lrs}) \in \mathbb{R}^L$, and $\mathcal{B}_1 \in \mathbb{R}^{L \times L}$ is a matrix given by

$$\mathcal{B}_1 = \int_0^1 \mathbf{B}(t)dt \int_0^1 \mathbf{B}^\top(t)dt = \begin{bmatrix} \big(\int_0^1 B_1(t)dt\big)^2 & \cdots & \int_0^1 B_1(t)dt \int_0^1 B_L(t)dt \\ \vdots & \ddots & \vdots \\ \int_0^1 B_L(t)dt \int_0^1 B_1(t)dt & \cdots & \big(\int_0^1 B_L(t)dt\big)^2 \end{bmatrix},$$

and

$$\boldsymbol{\Omega}_2(\mathcal{G}^{(c-1)}, \omega_\gamma^v) = \mathrm{diag}\Big(\frac{\dot{P}_{\omega_\gamma^v}(\|(a_{11}^{(c-1)})_v\|_2)}{\|(a_{11}^{(c-1)})_v\|_2}\mathcal{B}_2, \cdots, \frac{\dot{P}_{\omega_\gamma^v}(\|(a_{R1}^{(c-1)})_v\|_2)}{\|(a_{R1}^{(c-1)})_v\|_2}\mathcal{B}_2, \cdots, \frac{\dot{P}_{\omega_\gamma^v}(\|(a_{RS}^{(c-1)})_v\|_2)}{\|(a_{RS}^{(c-1)})_v\|_2}\mathcal{B}_2\Big)$$

is also an LRS-by-LRS block diagonal matrix, where $\|(a_{rs}^{(c-1)})_v\|_2 = \sqrt{\boldsymbol{\gamma}_{rs}^{(c-1)\top}\mathcal{B}_2\boldsymbol{\gamma}_{rs}^{(c-1)}}$ and $\mathcal{B}_2 \in \mathbb{R}^{L \times L}$ is a matrix given by

$$\begin{aligned}\mathcal{B}_2 &= \int_0^1 \mathbf{B}(t)\mathbf{B}^\top(t)dt - \int_0^1 \mathbf{B}(t)dt \int_0^1 \mathbf{B}^\top(t)dt \\ &= \begin{bmatrix} \int_0^1 B_1^2(t)dt - \big(\int_0^1 B_1(t)dt\big)^2 & \cdots & \int_0^1 B_1(t)B_L(t)dt - \int_0^1 B_1(t)dt \int_0^1 B_L(t)dt \\ \vdots & \ddots & \vdots \\ \int_0^1 B_L(t)B_1(t)dt - \int_0^1 B_L(t)dt \int_0^1 B_1(t)dt & \cdots & \int_0^1 B_L^2(t)dt - \big(\int_0^1 B_L(t)dt\big)^2 \end{bmatrix}.\end{aligned}$$

The vector $\mathrm{vec}(\mathcal{G}^{(c)})$ can be easily tensorized to produce the corresponding $\mathcal{G}^{(c)}$.

We iteratively update $\boldsymbol{\beta}$ and $\mathcal{G}$ until convergence, that is $\|\boldsymbol{\beta}^{(c)} - \boldsymbol{\beta}^{(c-1)}\|_2^2 + \|\mathcal{G}^{(c)} - \mathcal{G}^{(c-1)}\|_2^2 < 10^{-8}$ in this work. Finally, we set $\hat{\mathcal{G}}^\ddagger = \mathcal{G}^{(c)}$ and $\hat{\boldsymbol{\beta}}^\ddagger = \boldsymbol{\beta}^{(c)}$ upon convergence. We may obtain the penalized tensor function $\hat{\mathbf{A}}^\ddagger(t) = (\hat{a}_{rs}^\ddagger(t)) = (\mathbf{B}^\top(t)\hat{\boldsymbol{\gamma}}_{rs}^\ddagger)$.

Remark 2 Instead of shrinking $|(a_{rs})_c|$ and $\|(a_{rs})_v\|_2$ to zero, we could also operationally replace them by $|(\boldsymbol{\gamma}_{rs})_c|$ and $\|(\boldsymbol{\gamma}_{rs})_v\|_2$ where $(\boldsymbol{\gamma}_{rs})_c = \frac{1}{L}\sum_{l=1}^L \gamma_{lrs}$ is the mean and $(\boldsymbol{\gamma}_{rs})_v = \boldsymbol{\gamma}_{rs} - (\boldsymbol{\gamma}_{rs})_c \in \mathbb{R}^L$ represents the deviation function. These two pairs are quite similar in both theory and practice since we can choose the order of spline to ensure a one-to-one corresponding projection. More specifically, we can expand $|(\boldsymbol{\gamma}_{rs})_c|^2 = \frac{1}{L^2}\boldsymbol{\gamma}_{rs}^\top\mathbf{1}\mathbf{1}^\top\boldsymbol{\gamma}_{rs}$, and $\|(\boldsymbol{\gamma}_{rs})_v\|_2^2 = \boldsymbol{\gamma}_{rs}^\top(\mathbf{I} - \frac{1}{L}\mathbf{1}\mathbf{1}^\top)(\mathbf{I} - \frac{1}{L}\mathbf{1}\mathbf{1}^\top)^\top\boldsymbol{\gamma}_{rs} = \boldsymbol{\gamma}_{rs}^\top(\mathbf{I} - \frac{1}{L}\mathbf{1}\mathbf{1}^\top)\boldsymbol{\gamma}_{rs}$, where $\mathbf{1} \in \mathbb{R}^L$ is a vector of ones and $\mathbf{I} \in \mathbb{R}^{L \times L}$ is an identity matrix.

Remark 3 To identify constant coefficients in $\mathbf{A}(t)$, we can also use the term $\|\dot{a}_{rs}\|_2 = \sqrt{\boldsymbol{\gamma}_{rs}^\top\dot{\mathcal{B}}\boldsymbol{\gamma}_{rs}}$ to replace $\|(a_{rs})_v\|_2$, where $\dot{\mathcal{B}} = \{\int_0^1 \dot{\mathbf{B}}_l(t)^\top\dot{\mathbf{B}}_{l'}(t)dt : l, l' = 1, \cdots, L\} \in \mathbb{R}^{L \times L}$. The idea is to shrink the first derivative $\|\dot{a}_{rs}\|_2$ to zero, then $a_{rs}(t)$ is a constant function. This penalty term will produce a similar result as $\|(a_{rs})_v\|_2$ (Guo and Li (2022)).

Remark 4 In numerical studies, we have also used Lasso (Tibshirani (1996)) and MCP (Zhang (2010)) penalty functions in place of $P_\omega(\cdot)$ to show the versatility of our methodology. For Lasso and MCP penalty in our experiment, $\dot{P}_\omega(\theta) = \omega\mathrm{sign}(\theta)$ and $\dot{P}_\omega(\theta) = (\omega - \frac{|\theta|}{a_0'})_+\mathrm{sign}(\theta)$ respectively, where $a_0' = 3$.

Remark 5 We obtain uniform estimation rates for both $\hat{\boldsymbol{\beta}}^{\ddagger}$ and $\hat{a}^{\ddagger}_{rs}(t)$ of order $O_p\big(\max\{(\log n/n)^{2/5}, (\log n/n^{1-\tau})^{1/2}\}\big)$ in the Supplementary(c.f. Theorem 3 in Section C.2). Moreover, with appropriately vanishing tuning parameters and minimum signal strength, the penalized estimators achieve selection consistency: zero parametric coefficients, zero constant components, and zero varying parts are recovered with probability tending to one (c.f. Theorem 4 in Section C.2).

怎么选ω

3.2. *Tuning penalty parameters and model refinement.* After obtaining $\hat{\mathbf{A}}^{\ddagger}(t)$ and $\hat{\boldsymbol{\beta}}^{\ddagger}$, we can select variables (whether β_k can be shrunk to zero), and identify constant coefficients (whether $a_{rs}(t)$ is a zero constant function, non-zero constant function or varying function) simultaneously. To this end we need tune penalty parameters ω_β, ω^c_γ and ω^v_γ to achieve accurate results. We thus choose the optimal tuning parameters ω^*_β, $\omega^{c,*}_\gamma$ and $\omega^{v,*}_\gamma$ by a data-driven grid search method minimizing the Bayesian information criterion (BIC). Other criteria such as extended Bayesian information criterion (eBIC, Chen and Chen 2008) and generalised information criterion (GIC, Fan and Tang 2013) can also be adopted. More specifically, we select the best tuning parameters by minimizing

(29)
$$\mathrm{BIC}(\omega_\beta, \omega^c_\gamma, \omega^v_\gamma) = \log\Big(\frac{\sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \langle \mathcal{H}_{ij}(t_i), \hat{\mathcal{G}}^{\ddagger}\rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^{\ddagger}]^2}{2n}\Big) + (d_\beta + d_c + d_v L)\frac{\log(n)}{n},$$

where $\hat{\mathcal{G}}^{\ddagger}$ and $\hat{\boldsymbol{\beta}}^{\ddagger}$ are the solution of (22) given ω_β, ω^c_γ and ω^v_γ, d_c and d_v are the numbers of functions estimated as nonzero constant functions and varying functions respectively, and d_β is the number of non-zero coefficients in $\hat{\boldsymbol{\beta}}^{\ddagger}$. L is equivalent number of constant coefficients for a varying function, and exactly equal to the number of spline basis in (22).

4. Simulation study.

4.1. *Estimation consistency.* The data were generated according to (8), where t_i, $i = 1, \cdots, n$, are randomly sampled from the uniform distribution $U(0,1)$. We first consider the following two settings:

Case I : Each subject $\mathcal{X}_i$ is a $p_1 \times p_2$ tensor with order $D = 2$, $p_1 = p_2 = 80$ for $i = 1, \cdots, n$. Evenly divide $\mathcal{X}_i$ into $S = 16$ parts with $p'_1 = p'_2 = 20$, by partitioning it into $S^{(1)} = S^{(2)} = 4$ equal segments along length and width separately. Set $R = 10$. For each part $\tilde{\mathcal{X}}^{(s)}$ expressed in the form of (6), $\lambda^{(s)}_r = 1$, $\mathbf{x}^*_{rs} \in \mathbb{R}^n$ is a Gaussian ensemble vector whose entries are generated i.i.d. from the standard normal distribution, and any $\mathbf{x}^{(sd)}_r, \mathbf{x}^{(sd)}_{r'} \in \mathbb{R}^{p'_d}$, $r \neq r'$ are orthonormal and obtained from the QR decomposition of a random matrix. The tensor function $\mathcal{A}(t)$ has the same dimension of $\mathcal{X}_i$, and for each part, corresponding $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R a_{rs}(t)\mathbf{x}^{(s1)}_r \circ \mathbf{x}^{(s2)}_r$.

1.设置λ=1,
2. 设置X是Gauss,
这样所有n个样本都
被设置好

Case II : Each subject $\mathcal{X}_i$ is a $p_1 \times p_2 \times p_3$ tensor with order $D = 3$, $p_1 = p_2 = p_3 = 40$ for $i = 1, \cdots, n$. Evenly divide $\mathcal{X}_i$ into $S = 64$ parts by partitioning it into $S^{(1)} = S^{(2)} = S^{(3)} = 4$ equal segments along length, width, and height separately, with $p'_1 = p'_2 = p'_3 = 10$. Set $R = 5$. For each $\tilde{\mathcal{X}}^{(s)}$ in (6), $\lambda^{(s)}_r = 1$, and $\mathbf{x}^*_{rs} \in \mathbb{R}^n$, $\mathbf{x}^{(sd)}_r \in \mathbb{R}^{p'_d}$ are generated in the same way as Case I. Tensor function $\mathcal{A}(t)$ has the same dimension of $\mathcal{X}_i$, and for each part, corresponding $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R a_{rs}(t)\mathbf{x}^{(s1)}_r \circ \mathbf{x}^{(s2)}_r \circ \mathbf{x}^{(s3)}_r$.

In both cases, We consider 4 different functions: $a_{1rs}(t) = 4\sqrt{rs/RS}(t-0.5)^2$, $a_{2rs}(t) = \sqrt{rs/RS}t^{0.5}$, $a_{3rs}(t) = 1.75\sqrt{rs/RS}(\exp-(3t-1)^2 + \exp-(4t-3)^2 - 0.75)$, $a_{4rs}(t) = \sqrt{rs/RS}\sin(2\pi(t-0.5))$. We set $p_0 = 2$ and $\boldsymbol{\beta} = (3,3)^\top$. $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is a Gaussian ensemble matrix. The random error ϵ_i follows the standard normal distribution $N(0,1)$.

For each simulated data, we carry out the estimation procedure introduced in Section 2 to fit the model. After 500 simulations, we summarize the performance of our varying coefficient tensor regression (VCTR) model in Table 1 with sample size $n = 2000$ and 5000. The estimation errors for functions are integrated over the range of t. The errors for $\mathbf{A}(t)$ and $\boldsymbol{\beta}$ are averaged across all their components. These empirical results affirm the estimation consistency of the semi-parametric estimators for regression coefficients.

We also show the prediction errors of Case I and II with 10-fold cross validation in Figure 2. We compare our VCTR model with two constant coefficient tensor models, based on Miranda et al. (2018) and Zhou, Li and Zhu (2013), respectively. When the true model involves varying coefficients, our model clearly demonstrates an advantage over those existing models.

TABLE 1

Estimation results of Case I and II for 500 simulations. The accuracy of $\hat{\mathbf{A}}^{\dagger}(t)$ is measured by MIAE and RMISE, where MIAE $= \mathrm{E}\big[\int_0^1 |\hat{\mathbf{A}}^{\dagger}(t) - \mathbf{A}(t)|dt\big]$, and RMISE $= \mathrm{E}^{1/2}\big[\int_0^1 (\hat{\mathbf{A}}^{\dagger}(t) - \mathbf{A}(t))^2 dt\big]$. The accuracy of $\hat{\boldsymbol{\beta}}^{\dagger}$ is measured by MAE and MSE, where MAE $= \frac{1}{p_0}\sum_{j=1}^{p_0} |\hat{\beta}_j^{\dagger} - \beta_j|$, and RMSE $= \big[\frac{1}{p_0}\sum_{j=1}^{p_0} (\hat{\beta}_j^{\dagger} - \beta_j)^2\big]^{1/2}$.

验证sec2（无稀疏）方法：

随着n增大，各个A(t), β 的估计量的确会收敛；

Case	Function		$\hat{\mathbf{A}}^{\dagger}(t)$		$\hat{\boldsymbol{\beta}}^{\dagger}$	
			MIAE	RMISE	MAE	RMSE
I	$\mathbf{A}_1(t)$	2000	0.0622(0.0011)	0.0780(0.0012)	0.0343(0.0263)	0.0371(0.0267)
		5000	0.0272(0.0011)	0.0341(0.0013)	0.0200(0.0144)	0.0218(0.0145)
	$\mathbf{A}_2(t)$	2000	0.0636(0.0023)	0.0799(0.0029)	0.0455(0.0274)	0.0518(0.0302)
		5000	0.0279(0.0007)	0.0349(0.0009)	0.0270(0.0137)	0.0314(0.0158)
	$\mathbf{A}_3(t)$	2000	0.0648(0.0016)	0.0817(0.0021)	0.0441(0.0301)	0.0485(0.0337)
		5000	0.0398(0.0006)	0.0392(0.0008)	0.0259(0.0168)	0.0394(0.0154)
	$\mathbf{A}_4(t)$	2000	0.0678(0.0018)	0.0852(0.0021)	0.0340(0.0181)	0.0364(0.0179)
		5000	0.0358(0.0016)	0.0448(0.0019)	0.0152(0.0086)	0.0165(0.0084)
II	$\mathbf{A}_1(t)$	2000	0.0687(0.0031)	0.0865(0.0038)	0.0510(0.0254)	0.0592(0.0305)
		5000	0.0232(0.0009)	0.0293(0.0010)	0.0211(0.0084)	0.0217(0.0084)
	$\mathbf{A}_2(t)$	2000	0.0709(0.0022)	0.0889(0.0028)	0.0409(0.0188)	0.0461(0.0206)
		5000	0.0239(0.0007)	0.0299(0.0010)	0.0235(0.0101)	0.0244(0.0109)
	$\mathbf{A}_3(t)$	2000	0.0854(0.0055)	0.1103(0.0071)	0.0401(0.0381)	0.0425(0.0386)
		5000	0.0407(0.0009)	0.0535(0.0011)	0.0256(0.0152)	0.0286(0.0175)
	$\mathbf{A}_4(t)$	2000	0.1117(0.0049)	0.1430(0.0064)	0.0509(0.0245)	0.0541(0.0258)
		5000	0.0578(0.0012)	0.0727(0.0013)	0.0305(0.0162)	0.0362(0.0205)

验证sec3方法：

4.2. *Selection accuracy.* We now evaluate the accuracy of the penalization method for high-dimensional covariates introduced in Section 3. Let $\mathcal{F}_1 = \{(r,s) : \|(a_{rs})_v\|_2 > 0\}$, $\mathcal{F}_2 = \{(r,s) : |(a_{rs})_c| > 0, \|(a_{rs})_v\|_2 = 0\}$ and $\mathcal{F}_3 = \{(r,s) : |(a_{rs})_c| = 0, \|(a_{rs})_v\|_2 = 0\}$ be the index set of varying, constant non-zero and constant zero coefficients in tensor function $\mathbf{A}(t)$. Obviously, $\{(r,s) : r \in \{1,\cdots,R\}, s \in \{1,\cdots,S\}\} = \mathcal{F}_1 \cup \mathcal{F}_2 \cup \mathcal{F}_3$ and $\mathcal{F}_i \cap \mathcal{F}_j = \emptyset$ for $i,j \in \{1,2,3\}$. Also let $\mathcal{F}_4 = \{k, |\beta_k| > 0\}$ and $\mathcal{F}_5 = \{k, |\beta_k| = 0\}$ be the index set for significant and sparsed coefficients in one-way parameter $\boldsymbol{\beta}$. Similarly, we have $\{1,\cdots,p_0\} = \mathcal{F}_4 \cup \mathcal{F}_5$ and $\mathcal{F}_4 \cap \mathcal{F}_5 = \emptyset$. We denote $|\mathcal{F}|$ as the number of elements in a set $\mathcal{F}$. In this subsection, we denote sets $\hat{\mathcal{F}}_i$, $i = 1,\cdots,5$ as the estimator of ground truth $\mathcal{F}_i$.

F1-3: (r,s) 指标

F1非零函数

F2常数函数

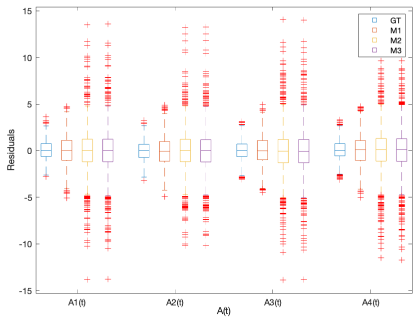
F3零函数

F4-5: k指标

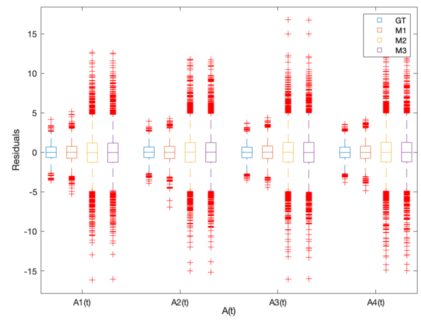
F4: β_k!=0

F5: β_k=0

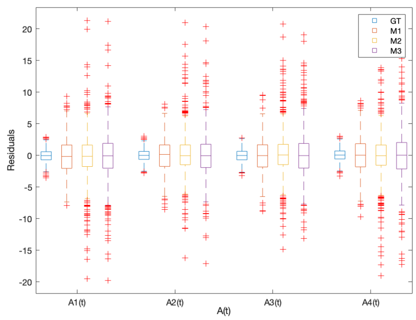
In Case III, we consider the selection performance under various spatial correlation among tensor covariates. In case IV, we consider tensor covariates with unknown coefficients greater than the sample size. In both cases, we adopt three types of familiar penalty functions: Lasso, SCAD, and MCP. The tuning parameters are chosen by BIC.



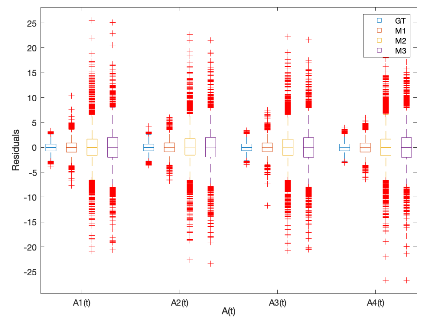
(a) Prediction error of Case I, n=2000.



(b) Prediction error of Case I, n=5000.



(c) Prediction error of Case II, n=2000.



(d) Prediction error of Case II, n=5000.

Fig 2: Box plots of prediction errors of Case I and II with 10-fold cross validation. Model M_1 is our VCTR model. Models M_2 and M_3 are constant coefficient tensor models, estimated by Miranda et al. (2018) and Zhou, Li and Zhu (2013), respectively.

Case III : Each subject $\mathcal{X}_i$ follows the same dimension as Case I. To allow the spatial correlation among different parts, for any two subject factor matrices $\mathbf{X}^{*(s)}$ in different partitions, entries have a first order auto-regressive covariance matrix with covariance $\rho^{|s^{(1)}-s'^{(1)}|+|s^{(2)}-s'^{(2)}|}$ for $1 \le s^{(1)}, s'^{(1)} \le S^{(1)}$ and $1 \le s^{(2)}, s'^{(2)} \le S^{(2)}$. Tensor function $\mathcal{A}(t)$ is designed similarly as Case I with $\mathcal{A}^{(s)}(t) = \sum_r^R a_{rs}(t)\mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$. If $1 \le r \le s \le 10$, $a_{rs}(t) = \sqrt{rs/RS}\sin(2\pi(t-0.5))$; if $1 \le s < r \le 10$, $a_{rs}(t) = \sqrt{rs/RS}$, and the rest of the tensor functions are all 0s.

Case I的高亮部分：由iid变成现在的AR

Case IV : Each subject $\mathcal{X}_i$ is an $p_1 \times p_2 \times p_3$ tensor with order $D = 3$, $p_1 = p_2 = p_3 = 80$ for $i = 1, \cdots, n$. Evenly divide $\mathcal{X}_i$ into $S = 64$ parts by partitioning it into $S^{(1)} = S^{(2)} = S^{(3)} = 4$ equal segments along length, width and height separately, with $p'_1 = p'_2 = p'_3 = 20$. Set $R = 20$. For each $\tilde{\mathcal{X}}^{(s)}$ in (6), $\lambda_r^{(s)} = 1$, and $\mathbf{x}_{rs}^* \in \mathbb{R}^n, \mathbf{x}_r^{(sd)} \in \mathbb{R}^{p'_d}$ are generated in the same way as Case II. Tensor function $\mathcal{A}(t)$ has the same shape as $\mathcal{X}$, and for each part, corresponding $\mathcal{A}^{(s)}(t) = \sum_r^R a_{rs}(t)\mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$. If $1 \le r \le s \le 10$, $a_{rs}(t) = \sqrt{rs/RS}\sin(2\pi(t-0.5))$; if $1 \le s < r \le 10$, $a_{rs}(t) = \sqrt{rs/RS}$, and the rest of the tensor functions are all 0s.

In both cases, We set $p_0 = 5$ and $\boldsymbol{\beta} = (1, 1, 0, 0, 0)^\top$. $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is a Gaussian ensemble matrix. The random error ϵ_i is generated in the same way as Case I and II.

The results for Case III are summarized in Table 2. The penalized estimation correctly identifies the true zero coefficients and non-zero coefficients with high probability. The covariate correlation slightly affects the accuracy where wrong selection occurs more often with higher auto-correlation. All three penalty functions perform quite well.

TABLE 2
Estimation results of 500 simulations for Case III. We use sensitivity (se), specificity (sp), positive predictive value (ppv), and negative predictive value (npv) to evaluate identification accuracy, where se $= \mathrm{P}(\mathcal{F}_i \cap \hat{\mathcal{F}}_i | \mathcal{F}_i)$*, ppv* $= \mathrm{P}(\mathcal{F}_i \cap \hat{\mathcal{F}}_i | \hat{\mathcal{F}}_i)$*, sp* $= \mathrm{P}(\mathcal{F}_i^c \cap \hat{\mathcal{F}}_i^c | \mathcal{F}_i^c)$ *and npv* $= \mathrm{P}(\mathcal{F}_i^c \cap \hat{\mathcal{F}}_i^c | \mathcal{F}_i^c)$*. Sample size* $n = 5000$*. Penalty parameters of Lasso, SCAD, and MCP are chosen by BIC.*

	Lasso			SCAD			MCP		
ρ	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
				Zero coefficient in $\mathbf{A}(t)$					
se	0.9965	0.9948	0.9926	0.9961	0.9952	0.9930	0.9974	0.9917	0.9913
ppv	0.9811	0.9849	0.9793	0.9922	0.9918	0.9827	0.9939	0.9917	0.9810
sp	0.9516	0.9615	0.9473	0.9802	0.9791	0.9560	0.9846	0.9791	0.9516
npv	0.9910	0.9865	0.9807	0.9901	0.9878	0.9820	0.9934	0.9792	0.9775
				Constant non-zero coefficient in $\mathbf{A}(t)$					
se	0.9712	0.9788	0.9558	0.9942	0.9827	0.9404	0.9885	0.9827	0.9481
ppv	0.9981	0.9923	0.9881	0.9868	0.9884	0.9781	0.9962	0.9848	0.9725
sp	0.9996	0.9985	0.9978	0.9974	0.9978	0.9959	0.9993	0.9970	0.9948
npv	0.9944	0.9959	0.9915	0.9989	0.9967	0.9886	0.9978	0.9967	0.9900
				Varying coefficient in $\mathbf{A}(t)$					
se	0.9256	0.9385	0.9359	0.9615	0.9744	0.9359	0.9795	0.9744	0.9333
ppv	0.9816	0.9787	0.9712	0.9948	0.9873	0.9463	0.9899	0.9720	0.9608
sp	0.9975	0.9972	0.9961	0.9993	0.9982	0.9925	0.9986	0.9961	0.9947
npv	0.9898	0.9915	0.9912	0.9947	0.9965	0.9911	0.9972	0.9964	0.9908
				Prediction error of $\hat{\mathbf{A}}^{\ddagger}(t)$					
MAE	0.0109	0.0111	0.0134	0.0114	0.0122	0.0127	0.0136	0.0175	0.0195
RMISE	0.0198	0.0217	0.0219	0.0192	0.0234	0.0240	0.0245	0.0397	0.0414
				Prediction error of $\hat{\boldsymbol{\beta}}^{\ddagger}$					
MAE	0.0047	0.0082	0.0092	0.0045	0.0063	0.0136	0.0046	0.0058	0.0141
RMSE	0.0077	0.0133	0.0149	0.0067	0.0101	0.0209	0.0080	0.0096	0.0207

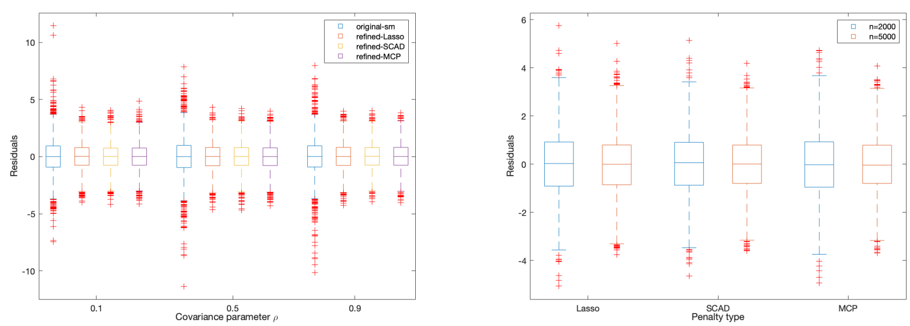
The results for Case IV are summarized in Table 3. We note that this case is more complicated than previous cases since the tensor coefficient has a greater dimension $RS = 20 \times 64$. The un-penalized estimation approach is infeasible for this challenging case. In general the selection accuracy improves with increasing sample size n while the three penalty functions perform similarly well. We plot prediction errors of Case III and IV in Figure 3.

5. Glaucoma management with fundus images. The scientific objective of the GRAPE study is to quantify the age-dependent relationship between IOP and fundus images of glaucoma patients, adjusting other clinical confounders. Figure 1 displays an example of fundus images (CFP and ROI), before being fed into our model estimation process. Following an initial data processing, we keep a set of $n = 591$ samples. We first resize all remaining images into $\mathcal{X}_i \in \mathbb{R}^{p_1 \times p_2 \times p_3}$, where $p_1 = 192$, $p_2 = 192$ and $p_3 = 3$, indicating length, width and number of color channels respectively. In this dataset, both oculus dexter and oculus sinister are available, in which the optic disc and macula are positioned oppositely and blood vessels may have different orientations. To avoid these discrepancies during the tensor partitioning process, we flipped all OS images horizontally and concatenate unified fundus images together as $\tilde{\mathcal{X}} \in \mathbb{R}^{n \times p_1 \times p_2 \times p_3}$. Then, an even division is applied to the length and the width, resulting with $S^{(1)}$ and $S^{(2)}$ equal segments and a total of $S = S^{(1)} \times S^{(2)}$ partitions $\tilde{\mathcal{X}}^{(s)}$. For each partition, a CP decomposition with rank R is applied. Focus on individual, the factor matrix of each subject can be obtained with $\mathbf{X}_i^* \in \mathbb{R}^{R \times S}, i = 1, \cdots, n$,

TABLE 3

Estimation results of 500 simulations for Case IV. We use sensitivity (se), specificity (sp), positive predictive value (ppv), and negative predictive value (npv) to evaluate identification accuracy, where se $= \mathrm{P}(\mathcal{F}_i \cap \hat{\mathcal{F}}_i | \mathcal{F}_i)$, *ppv* $= \mathrm{P}(\mathcal{F}_i \cap \hat{\mathcal{F}}_i | \hat{\mathcal{F}}_i)$, *sp* $= \mathrm{P}(\mathcal{F}_i^c \cap \hat{\mathcal{F}}_i^c | \hat{\mathcal{F}}_i^c)$ *and npv* $= \mathrm{P}(\mathcal{F}_i^c \cap \hat{\mathcal{F}}_i^c | \mathcal{F}_i^c)$. *All tuning parameters for Lasso, SCAD, and MCP are chosen by BIC.*

	Lasso			SCAD			MCP		
n	2000	3500	5000	2000	3500	5000	2000	3500	5000
Zero coefficient in $\mathbf{A}(t)$									
se	0.9916	0.9971	0.9975	0.9944	0.9962	0.9971	0.9913	0.9947	0.9947
ppv	0.9822	0.9891	0.9929	0.9954	0.9970	0.9982	0.9972	0.9967	0.9981
sp	0.6271	0.7729	0.8525	0.9051	0.9373	0.9627	0.9424	0.9322	0.9610
npv	0.7934	0.9295	0.9430	0.8879	0.9249	0.9423	0.8421	0.8750	0.8981
Constant non-zero coefficient in $\mathbf{A}(t)$									
se	0.6737	0.7895	0.8553	0.8974	0.9316	0.9658	0.9026	0.9263	0.9684
ppv	0.9217	0.9754	0.9739	0.9259	0.9332	0.9687	0.8989	0.9273	0.9497
sp	0.9981	0.9994	0.9993	0.9977	0.9979	0.9990	0.9968	0.9985	0.9977
npv	0.9901	0.9936	0.9956	0.9969	0.9979	0.9990	0.9970	0.9977	0.9990
Varying coefficient in $\mathbf{A}(t)$									
se	0.5143	0.7429	0.8476	0.9143	0.9476	0.9571	0.9048	0.9429	0.9476
ppv	0.5814	0.8564	0.8968	0.8265	0.8984	0.9129	0.7917	0.8161	0.8529
sp	0.9932	0.9978	0.9983	0.9967	0.9984	0.9982	0.9939	0.9963	0.9971
npv	0.9919	0.9957	0.9975	0.9986	0.9991	0.9993	0.9984	0.9990	0.9991
Prediction error of $\hat{\mathbf{A}}^{\ddagger}(t)$									
MAE	0.0242	0.0135	0.0108	0.0132	0.0055	0.0040	0.0097	0.0058	0.0041
RMISE	0.0375	0.0231	0.0194	0.0256	0.0140	0.0117	0.0223	0.0145	0.0115
Prediction error of $\hat{\boldsymbol{\beta}}^{\ddagger}$									
MAE	0.0361	0.0148	0.0090	0.0141	0.0073	0.0065	0.0218	0.0085	0.0064
RMSE	0.0514	0.0225	0.0133	0.0221	0.0124	0.0107	0.0325	0.0121	0.0101

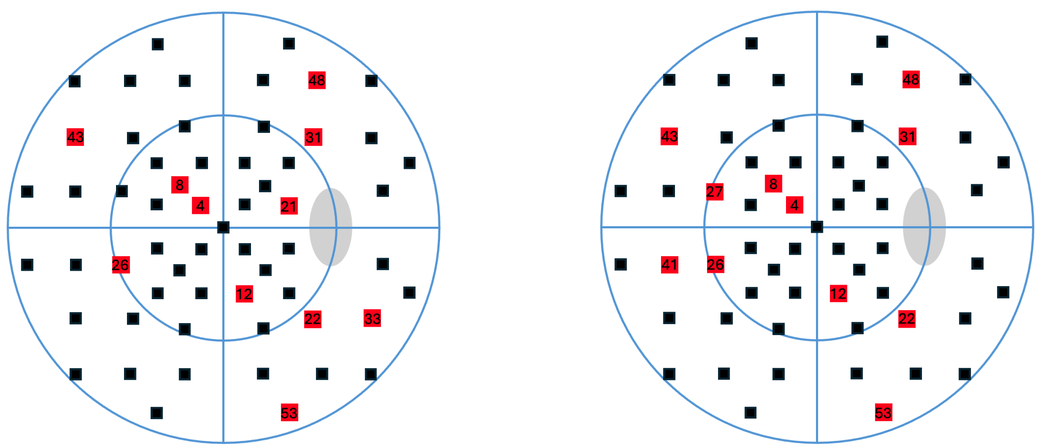


(a) Prediction error of Case III. (b) Prediction error of Case IV.

Fig 3: Box plots of prediction error of Case III and IV with 10-fold cross validation. (a) shows prediction errors of different covariance parameter ρ's in Case III, 'Oracle' is our tensor regression model with true model structure, 'Unpenalized' is the un-penalized tensor regression model (Section 2.3), and 'Lasso', 'SCAD', and 'MCP' are our final tensor regression model (Section 3.2) after model structure identification with Lasso, SCAD and MCP penalties (Section 3.1). (b) shows prediction errors of the final tensor regression model (Section 3.2) after model structure identification under overfitting situation (Case IV), with different sample size $n = 2000, 5000$. In case IV, un-penalized tensor regression model cannot work.

as we discuss in Section 2. We choose optic disc area as our ROI since the fundus manifestation of glaucoma contains optic disc rim narrowing, cup-to-disc ratio (CDR) increasing, large extent of parapapillary atrophy, and RNFL defect (Prum et al. (2016)), which mainly focused on the area around optic disc.

Our hypothesis is that the association between IOP and fundus images may vary with different ages, thus we consider model (3) with age being the index variable t_i. The one-way covariate $\mathbf{z}_i$ includes subject's gender and 59 view field values (cf. locations in Figure 1.(f)). To achieve consistency under the same location for OD and OS image, we record VF value along trajectory of scanning lines, as shown in (c) & (f) in Figure 1. All continuous variables are centered and standardized, and 10-fold cross validation is used to select S and R, and the best partition is $S = 3 \times 3 \times 1 = 9$ with the best CP rank $R = 2$ in both CFP and ROI images. See detailed cross validation results in the supplementary materials. With such specifications, we implement our VCTR model, and the estimated varying coefficient $\hat{\mathbf{A}}(t)$ and one-way constant coefficient $\hat{\boldsymbol{\beta}}$ are shown in Figure 5 and Table 4, respectively.



(a) Locations of retained VF with CFP.

(b) Locations of retained VF with ROI.

Fig 4: Fundus image analysis: Schematic diagrams of visual fields to illustrate the significant locations identified by penalized model. Red points in (a) and (b) are locations of retained $\hat{\boldsymbol{\beta}}^{\ddagger}$ with CFP and ROI image respectively. The index of VF is consistent with that in Figure 1(c).

From Table 4, insignificant gender effects are observed on CFP and ROI images. After model structure identification, two models retain similar set of VF locations, shown in Figure 4, but their spatial pattern differs. CFP based model concentrate around the optic disk area, reflecting its sensitivity to localized structure cues, while ROI based model retains a broader visual field area, compensating for global information beyond the optic disk area. VF locations (VF_{21}, VF_{22}, VF_{31}, VF_{33}) adjacent to the optic disc demonstrates significant negative associations with the response. Estimated coefficients of all VF locations are summarized in Table 4.

We compare the un-penalized estimator with the penalized estimator $\hat{a}_{rs}(t)$ in Figure 5. There are more varying coefficients for ROI than CFP, especially for partitions surrounding the optic disc. This suggests the increasing cup-to-disc ratio could lead to greater variability of image region effects on patients' intraocular pressure levels. These new insights would be difficult to uncover without applying the proposed tensor varying coefficient model.

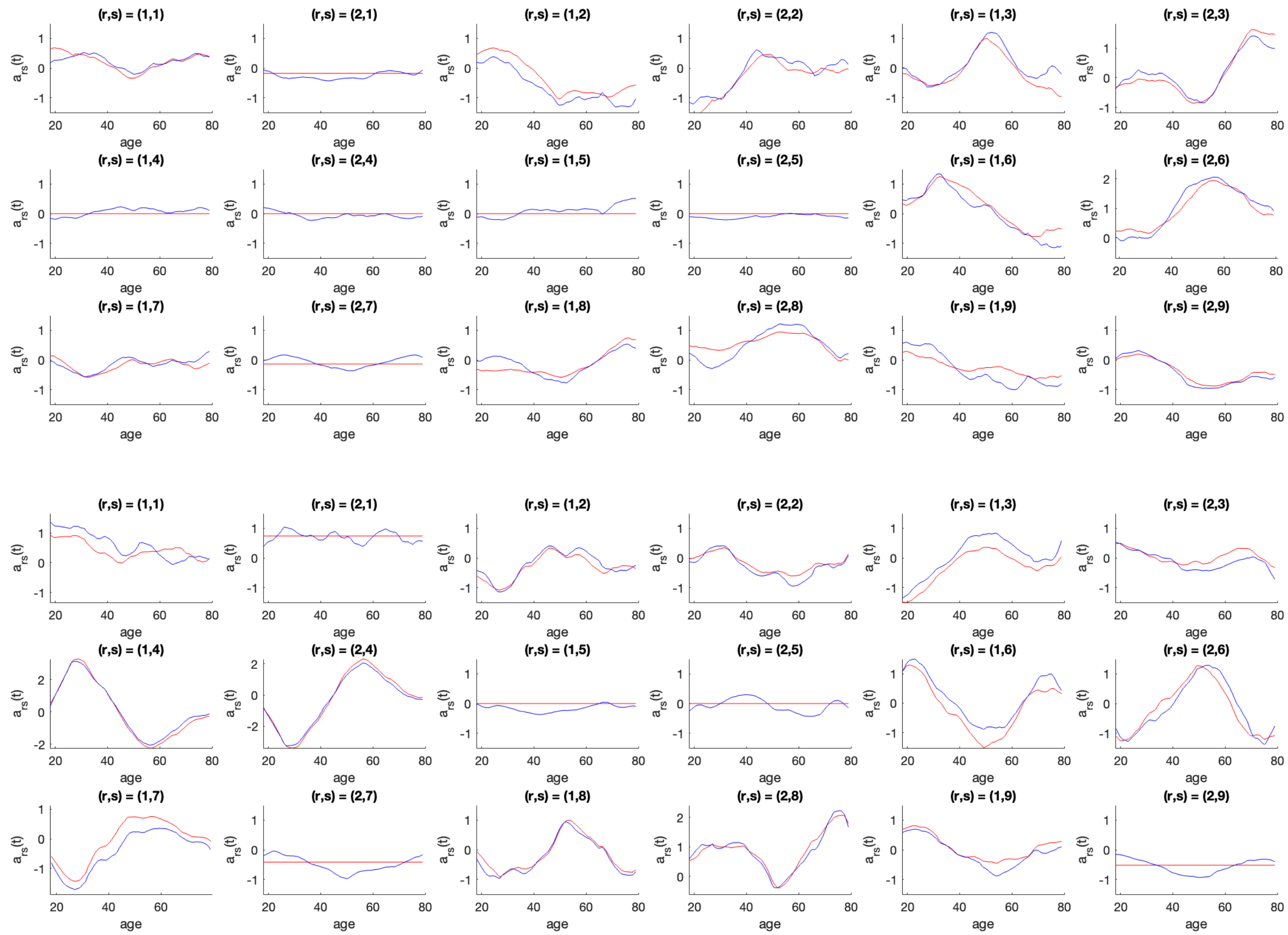


Fig 5: Fundus image analysis: estimated varying coefficient $\hat{a}_{rs}(t)$ for $\mathbf{X}_{\text{CFP}}$ generated from colored fundus photogragh (CFP, the top three rows) and $\mathbf{X}_{\text{ROI}}$ generated from region of interest (ROI, the bottom three rows). Blue lines are $\hat{a}_{rs}(t)$ from original kernel smoothing, and red lines are $\hat{a}_{rs}(t)$ from refined kernel smoothing after identifying coefficients.

6. Conclusion. Tensor regression techniques are increasingly recognized as powerful tools across a range of modern applications. Beyond the image data analysis presented in this study, these methods hold great potential in handling complex biomedical data structures, such as those arising in microbiome research or genetic network studies. In scenarios where intricate interaction effects are present, the varying coefficient tensor regression model proposed in this paper offers a flexible and effective framework. Its ability to accommodate high-dimensional and structured covariates makes it particularly well-suited for analyzing such complex data sources.

This study also has strong clinical relevance. The diagnostic insights derived from our analysis can assist ophthalmologists in making more accurate assessments of glaucoma status based on color fundus images, thereby enabling more precise and personalized treatment recommendations for patients. Currently, deep learning methods are widely adopted in ophthalmic image analysis for similar purposes. An exciting future direction would be to integrate our proposed model within a deep learning framework, which could enhance computational efficiency while producing interpretable and clinically actionable outputs.

SUPPLEMENTARY MATERIAL

Supplementary Material for "Identification of Age-dependent Effects of Fundus Images on Intraocular Pressure for Glaucoma Patients: A Tensor Regression Approach". Section A contains the additional simulation results. Section B contains the additional real data experiment results. In Section C, we present the statistical theory, showing the asymptotic distribution and consistency for estimated functions and coefficients. And Section D provides formal proofs for all theoretical results.

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TABLE 4

Fundus image analysis: Estimated constant coefficients $\hat{\boldsymbol{\beta}}$ for one-way covariate $\mathbf{z}$. $\hat{\boldsymbol{\beta}}^{\dagger}$ and $\hat{\boldsymbol{\beta}}^{\ddagger}$ are estimated from original kernel smoothing, and refined kernel smoothing after identifying coefficients. 95% confidence interval is generated from 500 bootstrap resamples.

$\mathbf{z}$	Colored fundus photograph (CFP)			Region of interest (ROI)		
	$\hat{\boldsymbol{\beta}}^{\dagger}$	$\hat{\boldsymbol{\beta}}^{\ddagger}$	95% CI	$\hat{\boldsymbol{\beta}}^{\dagger}$	$\hat{\boldsymbol{\beta}}^{\ddagger}$	95% CI
Gender	0.0449	0	/	-0.2304	0	/
VF_4	0.1350	-0.0251	[-0.0450, 0.2011]	0.0925	0.0590	[0.0165, 0.2468]
VF_8	0.0382	0.0246	[0.0025, 0.2254]	0.0433	0.1076	[0.0307, 0.2246]
VF_{12}	0.1360	0.0858	[-0.0659, 0.1524]	0.2076	0.1688	[0.0368, 0.2168]
VF_{21}	-0.1786	-0.1220	[-0.2059, 0.0164]	-0.0810	0	/
VF_{22}	-0.1801	-0.1390	[-0.2315, -0.0074]	-0.1262	-0.0927	[-0.1757, -0.0059]
VF_{26}	0.0326	0.0166	[0.0033, 0.1839]	-0.0591	0.0370	[-0.0879, 0.1060]
VF_{27}	-0.0171	0	/	-0.1407	-0.0613	[-0.1633, 0.0238]
VF_{31}	-0.1621	-0.1647	[-0.2598, -0.0630]	-0.0822	-0.0652	[-0.1113, 0.0648]
VF_{33}	0.0297	-0.0128	[-0.1200, 0.0677]	-0.0185	0	/
VF_{41}	0.0275	0	/	-0.0155	0.0075	[-0.0541, 0.1512]
VF_{43}	0.1662	0.0789	[0.0331, 0.1998]	0.1645	0.1033	[0.0594, 0.2230]
VF_{48}	-0.0805	-0.0516	[-0.1909, 0.0046]	-0.0539	-0.0148	[-0.2000, 0.0271]
VF_{53}	-0.0993	-0.0599	[-0.2186, -0.0403]	-0.0678	-0.0874	[-0.1744, -0.0158]

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